

1. List three data sources Power BI can connect to.

- **Microsoft Excel** – You can import data directly from Excel spreadsheets, including tables and pivot tables.
- **SQL Server Database** – Power BI can connect to on-premises or cloud-based SQL Server databases for robust data querying.
- **SharePoint Online** – Useful for accessing lists and files stored in SharePoint sites, especially in collaborative environments.

2. What is the first step to import data into Power BI Desktop?

- The first step to import data into Power BI Desktop is to **click on the “Get Data” button** located on the Home ribbon. This opens a window where you can choose from a wide variety of data sources such as Excel, SQL Server, SharePoint, and many others.

3. How do you refresh imported data in Power BI?

- **Open your Power BI report** – Make sure the report containing the imported data is open.
- **Click “Refresh” on the Home ribbon** – This triggers Power BI to re-query the data source and update the visuals with the latest data.
- **Verify the update** – Check your visuals and data tables to confirm that the new data has been loaded.

4. What file formats can Power BI import directly?(Name two.)

- **Excel (.xlsx, .xls)** – Ideal for importing structured data from spreadsheets and tables.
- **CSV (.csv)** – A widely used format for plain-text data with comma-separated values.

5. What does the “Navigator” window show after selecting a data source?

- **Available tables or sheets** from the selected data source (e.g., Excel sheets, database tables).
- **Preview pane** that displays a sample of the data in the selected table or sheet.
- **Checkboxes** to select one or multiple tables/sheets for loading or transforming.

6. Import Sales_Data.csv and load only the “Product” and “Price” columns.

Import and Filter Columns

1. **Open Power BI Desktop** Start a new report or open an existing one.

2. **Click “Get Data” > “Text/CSV”** Navigate to the location of Sales_Data.csv and click **Open**.
3. **Preview the file** Power BI will show a preview of the data. Click **Transform Data** to open Power Query Editor.
4. **Remove unwanted columns**
 - a. In Power Query, locate the **"Product"** and **"Price"** columns.
 - b. Select all other columns, right-click, and choose **Remove Columns**.
5. **Apply changes** Click **Close & Apply** to load the filtered data into Power BI.

7. How would you change OrderDate to a date format during import?

- **Import your data** Use **Get Data** to load your source (e.g., Excel, CSV, SQL).
- **Open Power Query Editor** Click **Transform Data** to access the editor.
- **Locate the OrderDate column** Find the column in the data preview pane.
- **Change the data type**
 - Select the **OrderDate** column.
 - Go to the **Transform** tab.
 - Click **Data Type** dropdown and choose **Date**.
- **Apply changes** Click **Close & Apply** to load the transformed data into Power BI.

8. What is the difference between “Load” and “Transform Data” in the import dialog?

Load

- **Directly imports** the selected data into Power BI.
- No changes or cleaning are applied.
- Best for clean, ready-to-use datasets.

Transform Data

- Opens the **Power Query Editor**.
- Lets you **clean, shape, and modify** the data before loading.
- Useful for filtering columns, changing data types, removing duplicates, and more.

In short:

- Use **Load** when your data is good to go.
- Use **Transform Data** when you need to prep it first.

9. Why might you see an error when connecting to SQL database?(Name one reason)

One common reason you might see an error when connecting to a SQL database in Power BI is **incorrect server or database credentials**. If the username, password, or server name is wrong or missing, Power BI won't be able to establish a connection.

10.How do you replace a data source after importing it?

Steps to Replace a Data Source

1. **Open Power Query Editor** Go to the **Home** tab and click **Transform Data**.
2. **Select the query tied to the old source** In the Queries pane on the left, click the query you want to update.
3. **Edit the Source step**
 - a. In the **Applied Steps** pane, find the step labeled **Source**.
 - b. Click the gear icon  next to it.
4. **Update the connection details**
 - a. For files: Browse to the new file (e.g., updated CSV or Excel).
 - b. For databases: Enter the new server or database name.
5. **Confirm and apply changes** Click **OK**, then choose **Close & Apply** to refresh your report with the new data source.

12.How would you change the data source if Sales_Data.csv changed?

Update the Data Source in Power BI Desktop

1. **Open Power Query Editor** Go to the **Home** tab and click **Transform Data**.
2. **Find the query for Sales_Data.csv** In the Queries pane, select the query that references the original CSV file.
3. **Edit the Source step**
 - a. In the **Applied Steps** pane, click the gear icon  next to **Source**.
 - b. A dialog box will open showing the current file path.
4. **Browse to the new file**
 - a. Click **Browse** and select the updated or relocated **Sales_Data.csv**.
 - b. Confirm and click **OK**.

5. Review and apply changes

- a. Check that the data loads correctly.
- b. Click **Close & Apply** to refresh your report with the new data.

13. Troubleshoot: Your CSV import fails due to a “mixed data type” error – how do you fix it?

Fixing Mixed Data Type Errors in Power BI

1. **Open Power Query Editor** Click **Transform Data** to access the editor.
2. **Identify the problematic column** Look for columns with a warning icon or unexpected values.
3. **Convert to a consistent data type**
 - a. Select the column.
 - b. Go to the **Transform** tab.
 - c. Choose a suitable data type (e.g., **Text**, **Decimal Number**, **Date**).
4. **Handle inconsistent values**
 - a. Use **Replace Errors** to substitute invalid entries.
 - b. Or use **Remove Rows > Remove Errors** to exclude them entirely.
5. **Apply changes** Click **Close & Apply** to reload the cleaned data into Power BI.

15. How would you automate data imports using Power BI and Power Automate?

Step-by-Step: Automate Sales_Data Import

1. Store Sales_Data in a Cloud Location

- Upload your Sales_Data.csv to a cloud service like **OneDrive**, **SharePoint**, or **Azure Blob Storage**.
- This ensures Power Automate can access it reliably.

2. Create a Power Automate Flow

- Go to Power Automate.
- Choose **Create > Automated Cloud Flow**.
- Set a trigger like:
 - **“When a file is created or modified”** in OneDrive or SharePoint.
 - Or **schedule** it to run daily/hourly.

3. Add Power BI Action

- After the trigger, add the action: **“Refresh a dataset”** (Power BI connector).
- Select your workspace and the dataset linked to your Power BI report.

4. Save and Test the Flow

- Save your flow and test it by updating or replacing `Sales_Data.csv`.
- Power BI will automatically refresh the dataset when the flow runs.